

Practice Quiz: Systems

Part A: Multiple Choice

1. A cortical column can be understood as: A) A complete brain region responsible for a single cognitive function B) A local inference engine maintaining predictions about specific sensory features C) A random grouping of neurons with no functional significance D) A structure found only in the visual cortex
2. The Markov Blanket of the brain is best identified with: A) The skull and meninges B) The sensory and motor surfaces of the body C) The blood-brain barrier D) The corpus callosum connecting the two hemispheres
3. Resting-state fMRI activity reflects: A) Neural noise with no functional significance B) The brain's ongoing generative model — prior predictions maintained even without external stimulation C) Damage to the default mode network D) The brain entering a sleep-like state
4. The salience network, anchored by the anterior insula and dACC, likely implements: A) Long-term memory storage B) Precision weighting — determining which prediction errors are amplified C) Motor planning and execution D) Language comprehension
5. Anosognosia (unawareness of deficit) is best explained in Active Inference as: A) Psychological denial as a defense mechanism B) A failure to update the generative model's self-component despite contradictory sensory evidence C) Damage to the visual cortex preventing visual feedback D) An emotional response to trauma
6. The mismatch negativity (MMN) in EEG experiments is considered a neural correlate of: A) Motor intention B) Prediction error — the brain's response to a stimulus that violates expected patterns C) Memory consolidation D) Emotional processing
7. The hierarchical organization of cortex (primary → association areas) maps onto: A) The temporal sequence of neural development B) The hierarchical depth of the

generative model (concrete → abstract predictions) C) The physical distance from the brainstem D) The density of myelinated axons

Part B: Short Answer

1. Explain why the default mode network's activity during rest may represent the brain's self-model rather than mere neural idling. What experimental evidence supports this interpretation?
2. A patient with depersonalization reports feeling detached from their own body — as if observing themselves from outside. Analyze this experience using Active Inference: what has happened to the precision of self-model predictions?
3. Design a simple experiment that could test the Active Inference prediction that mismatch negativity (MMN) amplitude correlates with the precision of auditory prediction errors.